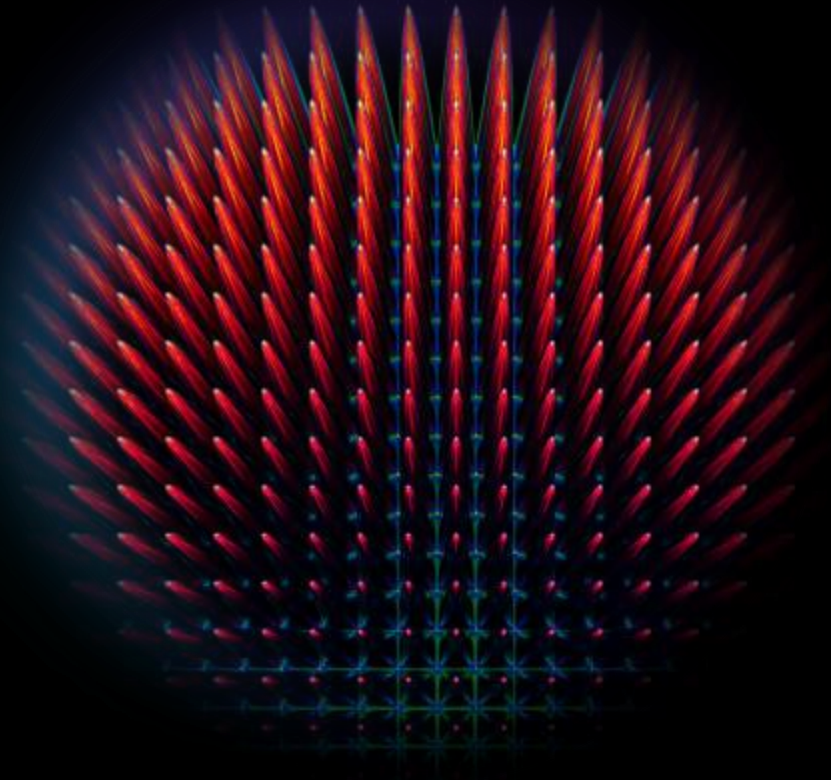


# Analysis of Machine Learning and Deep Learning Models for Retinal Disease Detection



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# Abstract

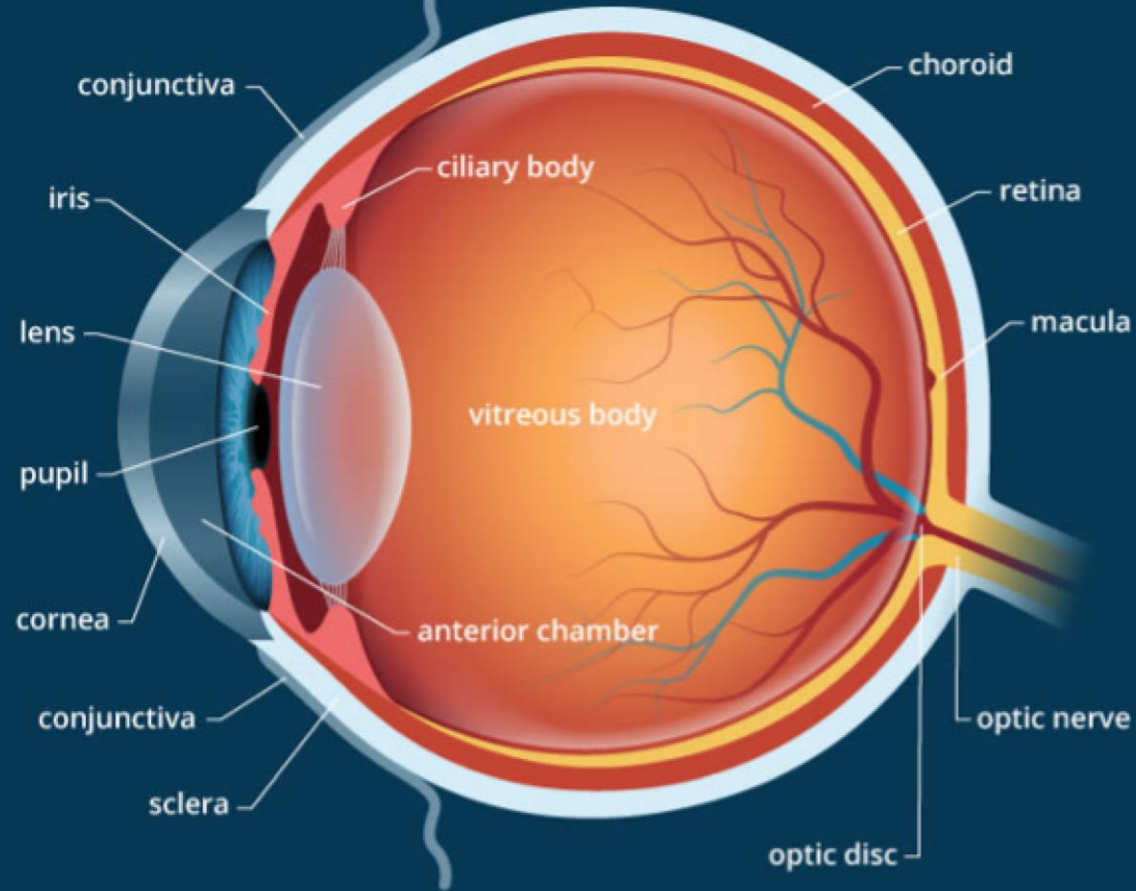
Retinal diseases represent a major public health concern, Early and accurate screening is essential to prevent vision loss, but conventional methods are time-consuming, resource-intensive, and require highly trained personnel who may not be readily available in remote regions. By training machine learning (ML) and deep learning (DL) models on color fundus images annotated by experts, automated systems can rapidly screen and classify various retinal diseases with high precision.

In this project, we apply and compare a range of ML and DL algorithms to the same dataset, aiming to demonstrate why deep learning models typically yield superior results in image-based medical tasks. Ultimately, our approach has the potential to make retinal screening more efficient, scalable, and accessible, particularly in underserved communities.

Keywords: Machine Learning (ML), Deep Learning (DL), Retinal Diseases, Medical image analysis, Feature Extraction, Retinopathy.



## Eye Anatomy



# Introduction

The retina is a thin layer at the back of the eye. It is made up of millions of tiny cells called rods and cones. These cells help us see by catching the light that comes into our eyes. The retina then sends this information to the brain through the optic nerve. This is how we see everything around us.

Some of the main retinal diseases we are focusing on in this project are Cataract, Diabetic Retinopathy, and Glaucoma. All these problems can make it hard to see and may even lead to blindness if they are not treated.

# What's Fundus?

To help doctors detect eye diseases, we use fundus images—detailed photos of the back of the eye, including the retina, macula, and optic disc. Taken with a special fundus camera, these images reveal signs of damage or disease.

In this project, we use fundus images to train machine learning (ML) and deep learning (DL) models to automatically identify conditions such as cataract, glaucoma, and diabetic retinopathy including normal fundus too.



Normal Fundus

# Literature Review

| Reference                   | Paper Title                                                                                                                                      | Result                                                                                                                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (Rahman et. al., 2024)      | Machine Learning and Deep Learning-based Approach in Smart Healthcare: Recent Advances, Applications, Challenges, and Opportunities.             | Deep learning, especially CNNs, is highly effective in medical image analysis, often outperforming traditional machine learning in classification accuracy.                            |
| (Kumar A. and Kaur 2024)    | Machine Learning and Deep Learning Based Healthcare System: A Review.                                                                            | The study compares methods and finds deep learning excels at medical image analysis by automatically extracting features, reducing manual feature engineering.                         |
| (Karabeg et. al., 2024)     | A pilot cost-analysis study comparing AI-based EyeArt® and ophthalmologist assessment of diabetic retinopathy in minority women in Oslo, Norway. | This pilot study in Oslo compares manual and AI grading for diabetic retinopathy, introducing Norway's first AI application for DR detection.                                          |
| (Khalil et. al. 2014)       | Review of machine learning techniques for glaucoma detection and prediction.                                                                     | A survey to detect glaucoma changes through fundus images. Preprocessing, feature extraction, feature selection, and machine learning (ML) techniques were discussed.                  |
| Gupta and Karandikar (2015) | Diagnosis of Diabetic Retinopathy using Machine Learning.                                                                                        | a survey study that analyses the automated techniques for DR diagnosis. A comparison was made between the algorithm that detects the various structural changes through fundus images. |



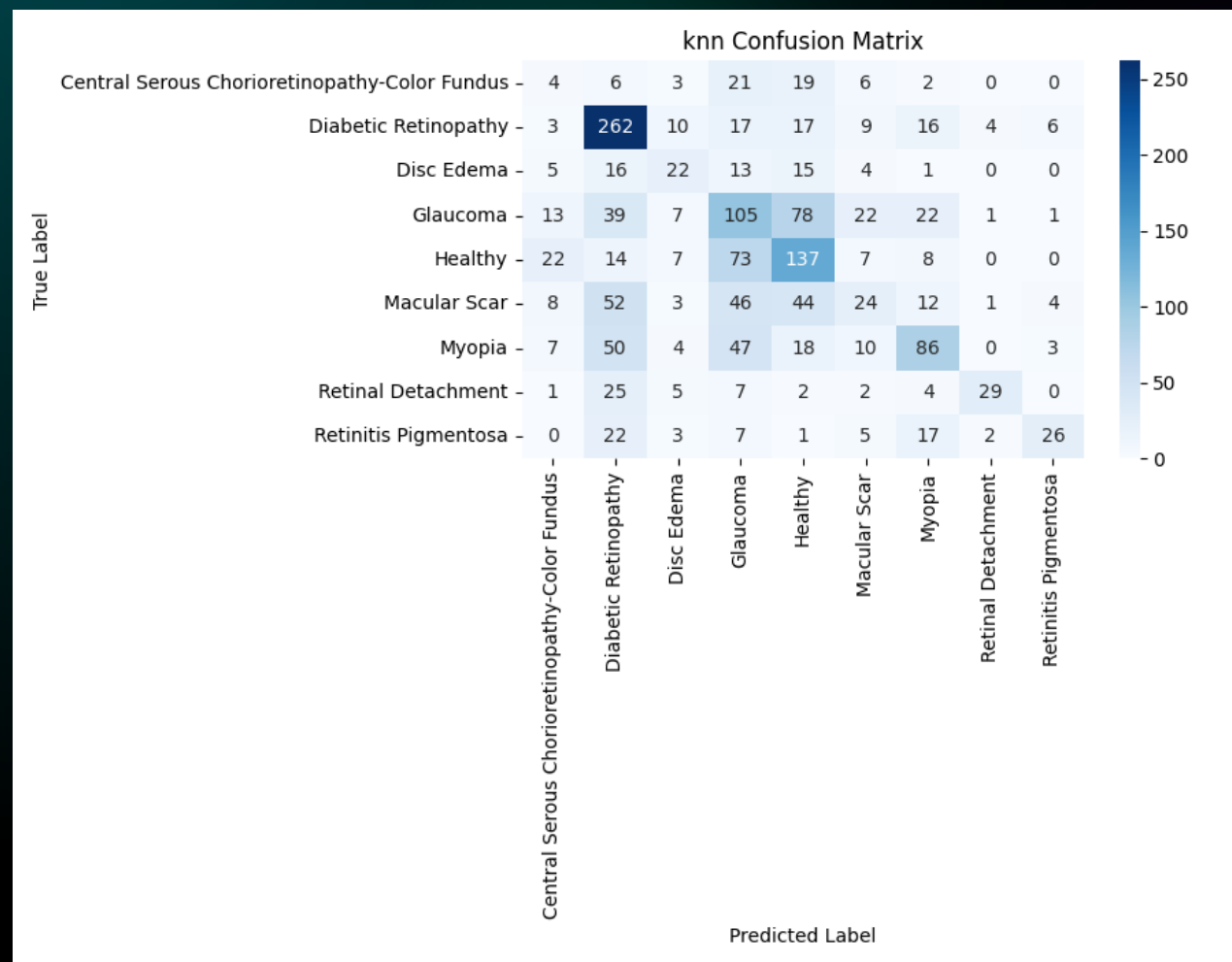
# About the Dataset

This dataset contains retinal images categorized into four primary classes: Normal, Diabetic Retinopathy, Cataract, and Glaucoma, with each class comprising approximately 1000 images. These images are curated from various reputable sources, such as IDRiD, Oculus Recognition, and HRF. It is organized into pre-labeled folders, with each folder dedicated to a specific class (retinal disease), streamlining the data preprocessing and loading process for model training.

| SL. NO. | Name of the Retinal disease | No. of Images | Percentage (%) |
|---------|-----------------------------|---------------|----------------|
| 1       | (N) Normal Fundus           | 1074          | 24.61          |
| 2       | (DR) Diabetic Retinopathy   | 1098          | 26.04          |
| 3       | (C) Cataract                | 1038          | 24.61          |
| 4       | (G) Glaucoma                | 1007          | 23.88          |

Although image quality is not a major issue, the quantity of images is crucial. In previous attempts, I used images from multiple classes with significantly different numbers of samples. This imbalance negatively affects the training of ML and DL models. As a result, the model may perform poorly, struggling to differentiate between classes because it has not been trained on an equal number of samples per class, or some classes have larger samples of data, but some have smaller samples of data. This poor performance becomes evident when analysing the model's results using a confusion matrix, which we can observe below.

# Why this dataset over others?

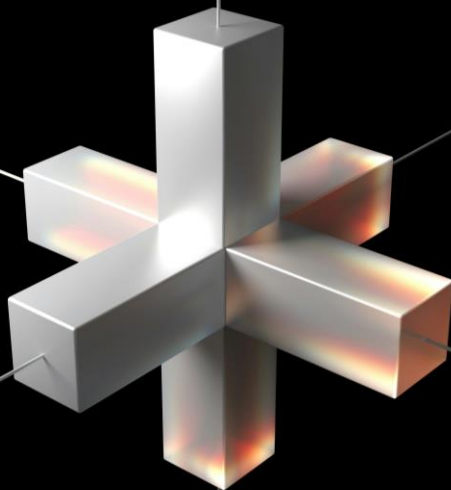


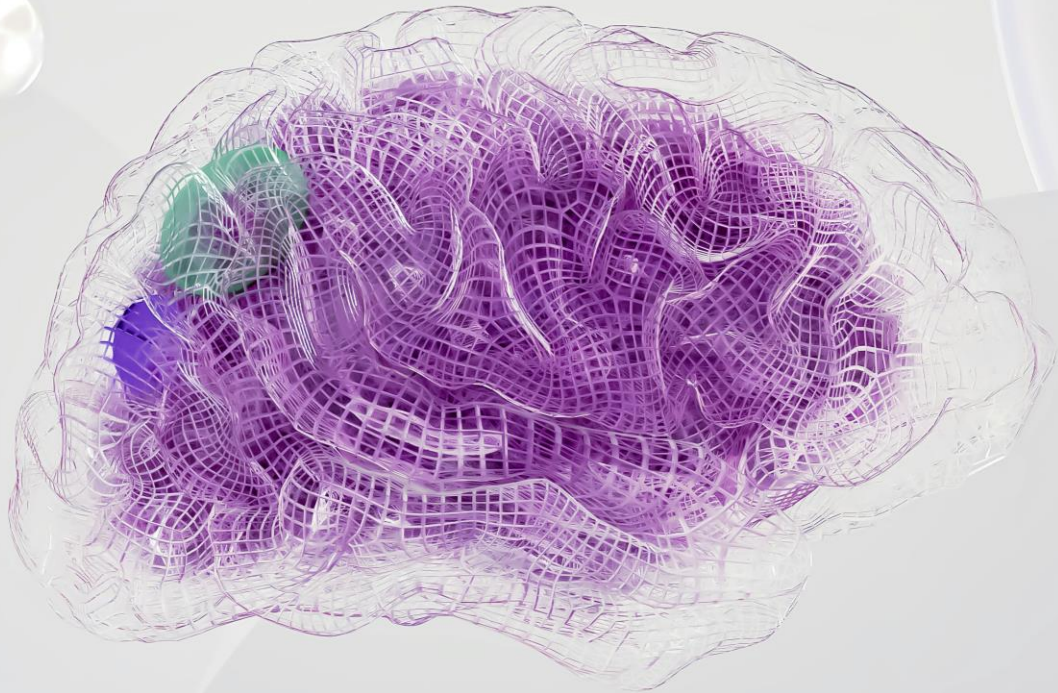




# Algorithms and Models

In this project, our goal is to build a system that can automatically identify and classify different retinal diseases from fundus images. To do this, we apply a variety of classification techniques, which fall mainly under two major categories: Machine Learning (ML) algorithms and Deep Learning (DL) models. Each of these techniques has its strengths and a unique way of working with data. By testing and analyzing both, we can get a clear idea about which approach is most effective for medical image classification, specifically in the context of retinal disease detection.





# Machine Learning Algorithms

## KNN

K-Nearest Neighbors classifies data points based on the majority label of their nearest neighbors.

## Decision Tree

Decision Tree splits data into branches using feature-based questions, creating an interpretable model structure.

## Random Forest

Random Forest combines multiple decision trees, improving accuracy and reducing overfitting in classification tasks.

# KNN

| Class                | Accuracy | Precision | Recall | Specificity | F1_score | FPR   | FNR   |
|----------------------|----------|-----------|--------|-------------|----------|-------|-------|
| Cataract             | 91.23    | 77.69     | 90.38  | 91.51       | 83.56    | 8.49  | 9.62  |
| Diabetic Retinopathy | 81.99    | 64.41     | 69.09  | 86.54       | 66.67    | 13.46 | 30.91 |
| Glaucoma             | 85.31    | 74.07     | 59.41  | 93.46       | 65.93    | 6.54  | 40.59 |
| Normal               | 81.28    | 63.73     | 60.75  | 88.25       | 62.2     | 11.75 | 39.25 |
| Macro Average        | 84.95    | 69.97     | 69.91  | 89.94       | 69.59    | 10.06 | 30.09 |



# Decision Tree

| Class                | Accuracy | Precision | Recall | Specificity | F1_score | FPR   | FNR   |
|----------------------|----------|-----------|--------|-------------|----------|-------|-------|
| Cataract             | 84.36    | 68.27     | 68.27  | 89.62       | 68.27    | 10.38 | 31.73 |
| Diabetic Retinopathy | 77.25    | 56.25     | 57.27  | 84.29       | 56.76    | 15.71 | 42.73 |
| Glaucoma             | 75.59    | 49.04     | 50.5   | 83.49       | 49.76    | 16.51 | 49.5  |
| Normal               | 72.27    | 45.1      | 42.99  | 82.22       | 44.02    | 17.78 | 57.01 |
| Macro Average        | 77.37    | 54.66     | 54.76  | 84.91       | 54.7     | 15.09 | 45.24 |



# Random Forest

| Class                | Accuracy | Precision | Recall | Specificity | F1_score | FPR   | FNR   |
|----------------------|----------|-----------|--------|-------------|----------|-------|-------|
| Cataract             | 90.52    | 78.57     | 84.62  | 92.45       | 81.48    | 7.55  | 15.38 |
| Diabetic Retinopathy | 86.02    | 75.76     | 68.18  | 92.31       | 71.77    | 7.69  | 31.82 |
| Glaucoma             | 81.04    | 60        | 62.38  | 86.92       | 61.17    | 13.08 | 37.62 |
| Normal               | 79.38    | 59.43     | 58.88  | 86.35       | 59.15    | 13.65 | 41.12 |
| Macro Average        | 84.24    | 68.44     | 68.51  | 89.51       | 68.39    | 10.49 | 31.49 |



# Deep Learning Algorithms

## ResNet101

A deep (CNN) with 101 layers, using residual connections for training stability.

## ResNet50

A 50-layer (CNN) employs residual connections to improve image classification performance.

## DenseNet121

Connects each layer to every other, enhancing feature reuse and efficient gradient flow.

# ResNet101

| Class                | Accuracy | Precision | Recall | Specificity | F1_score | FPR  | FNR   |
|----------------------|----------|-----------|--------|-------------|----------|------|-------|
| Cataract             | 95.27    | 87.3      | 96.49  | 94.82       | 91.67    | 5.18 | 3.51  |
| Diabetic Retinopathy | 100      | 100       | 100    | 100         | 100      | 0    | 0     |
| Glaucoma             | 93.62    | 87.34     | 80.23  | 97.03       | 83.64    | 2.97 | 19.77 |
| Normal               | 95.51    | 93.46     | 89.29  | 97.75       | 91.32    | 2.25 | 10.71 |
| Macro Average        | 96.1     | 92.03     | 91.5   | 97.4        | 91.66    | 2.6  | 8.5   |



# ResNet50

| Class                | Accuracy | Precision | Recall | Specificity | F1_score | FPR  | FNR   |
|----------------------|----------|-----------|--------|-------------|----------|------|-------|
| Cataract             | 95.74    | 95.28     | 88.6   | 98.38       | 91.82    | 1.62 | 11.4  |
| Diabetic Retinopathy | 99.76    | 99.11     | 100    | 99.68       | 99.55    | 0.32 | 0     |
| Glaucoma             | 94.33    | 84.44     | 88.37  | 95.85       | 86.36    | 4.15 | 11.63 |
| Normal               | 95.04    | 89.57     | 91.96  | 96.14       | 90.75    | 3.86 | 8.04  |
| Macro Average        | 96.22    | 92.1      | 92.23  | 97.51       | 92.12    | 2.49 | 7.77  |

# DenseNet121

| Class                | Accuracy | Precision | Recall | Specificity | F1_score | FPR  | FNR   |
|----------------------|----------|-----------|--------|-------------|----------|------|-------|
| Cataract             | 92.43    | 81.06     | 93.86  | 91.91       | 86.99    | 8.09 | 6.14  |
| Diabetic Retinopathy | 100      | 100       | 100    | 100         | 100      | 0    | 0     |
| Glaucoma             | 91.96    | 87.14     | 70.93  | 97.33       | 78.21    | 2.67 | 29.07 |
| Normal               | 94.33    | 90        | 88.39  | 96.46       | 89.19    | 3.54 | 11.61 |
| Macro Average        | 94.68    | 89.55     | 88.3   | 96.43       | 88.6     | 3.57 | 11.7  |

# Machine Learning overall performance Macro Average.

| Model         | Accuracy | Precision | Recall | Specificity | F1_score | FPR   | FNR   |
|---------------|----------|-----------|--------|-------------|----------|-------|-------|
| Decision Tree | 77.37    | 54.66     | 54.76  | 84.91       | 54.7     | 15.09 | 45.24 |
| Random Forest | 84.24    | 68.44     | 68.51  | 89.51       | 68.39    | 10.49 | 31.49 |
| KNN           | 84.95    | 69.97     | 69.91  | 89.94       | 69.59    | 10.06 | 30.09 |

# Deep Learning overall performance Macro Average.

| Model        | Accuracy | Precision | Recall | Specificity | F1_score | FPR  | FNR  |
|--------------|----------|-----------|--------|-------------|----------|------|------|
| DenseNet 121 | 94.68    | 89.55     | 88.3   | 96.43       | 88.6     | 3.57 | 11.7 |
| ResNet101    | 96.1     | 92.03     | 91.5   | 97.4        | 91.66    | 2.6  | 8.5  |
| ResNet50     | 96.22    | 92.1      | 92.23  | 97.51       | 92.12    | 2.49 | 7.77 |



# Comparing top Machine Learning and Deep Learning algorithms by overall performance.

Comparing traditional machine learning algorithms and deep learning models for retinal disease detection reveals deep learning's superior performance. While KNN achieved 84.95% accuracy, deep learning models performed better, with ResNet50 reaching the highest overall accuracy of 96.22%.

| Model    | Accuracy | Precision | Recall | Specificity | F1_score | FPR   | FNR   |
|----------|----------|-----------|--------|-------------|----------|-------|-------|
| KNN      | 84.95    | 69.97     | 69.91  | 89.94       | 69.59    | 10.06 | 30.09 |
| ResNet50 | 96.22    | 92.1      | 92.23  | 97.51       | 92.12    | 2.49  | 7.77  |

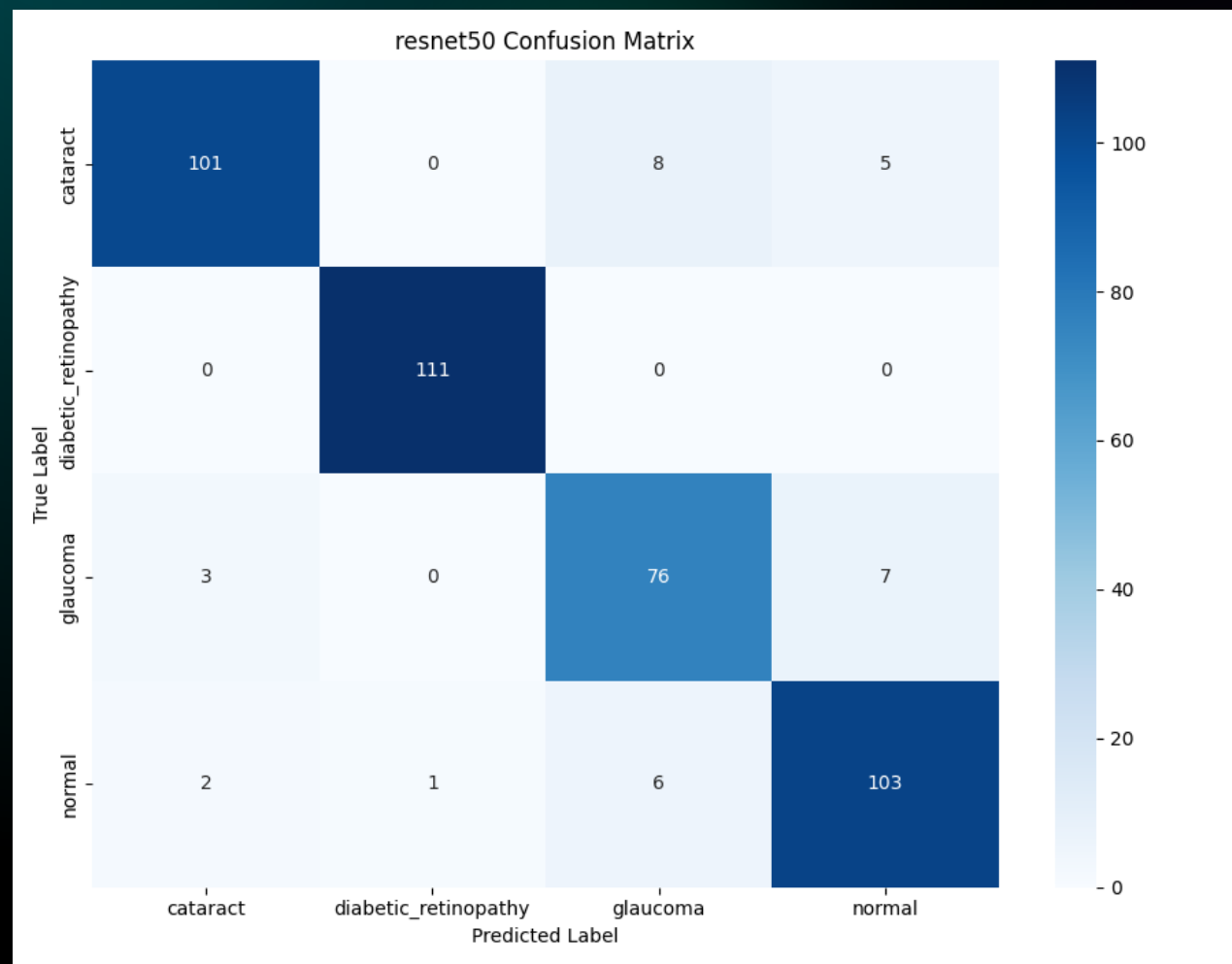


# Conclusion

The accuracy of ResNet50 (96.22%) is approximately 1.13 times higher than KNN (84.95%). Precision improves from 69.97% (KNN) to 92.1% (ResNet50), which is about 1.32 times higher. Recall rises from 69.91% (KNN) to 92.23% (ResNet50)—about 1.32 times better. Specificity increases from 89.94% to 97.51%, or about 1.08 times higher. The F1-score sees a significant improvement, from 69.59% to 92.12%, making it roughly 1.32 times higher. Notably, ResNet50 reduces the False Positive Rate (FPR) from 10.06% to 2.49%, a reduction by about 4 times, and the False Negative Rate (FNR) from 30.09% to 7.77%, which is nearly 3.87 times lower.



# Confusion Matrix of ResNet 50







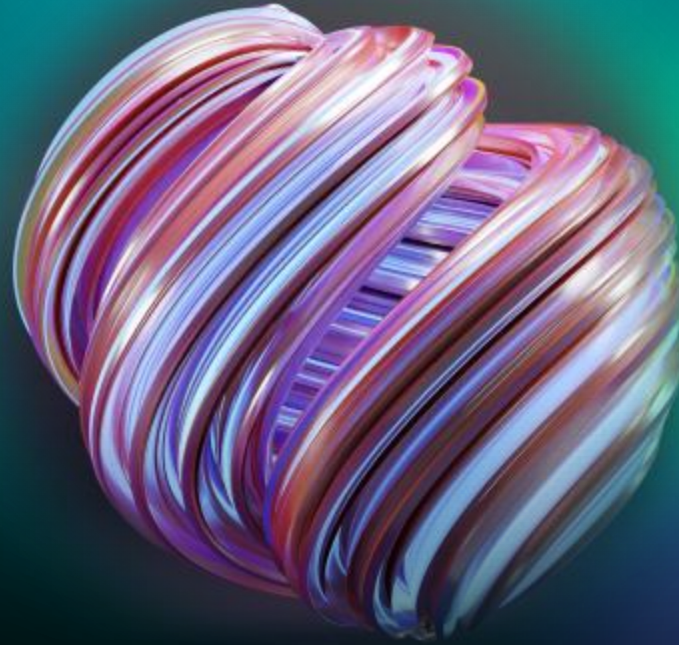
# Future work

Future research should expand datasets to diverse populations and integrate multi-modal data, such as medical history and genetics. Additionally, practical steps include developing user-friendly applications, conducting clinical validation, and training healthcare professionals to enhance remote retinal disease detection.

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**Thank you**

[github.com/nayaksomkar/RetinaDx](https://github.com/nayaksomkar/RetinaDx)